Suraj Samanta

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Education

Dr. Akhilesh Das Gupta Institute of Professional Studies (ADGIPS) — New Delhi, India
Bachelor of Technology in Computer Science Engineering | CGPA: 9.3 / 10.0 — *Aug. 2024 – May 2028 (Expected)*

Awards & Achievements

- **Winner — Elastic × Google Cloud Hacknight 2026:** Built *Vayu Watch*, a live Delhi air quality platform with DPCC/CPCB data, Gemini advisories, and Elasticsearch-indexed real-time history.
- **National Finalist — Smart India Hackathon 2025:** Built *SAHAI*, a full-stack mental health platform with WebSocket chat and RAG-based AI therapy (Pinecone + FastAPI), selected from thousands of teams.
- **Top 5 in India — NASA Space Apps Challenge 2025 (823 teams):** Built *ORBITAIR*, an AQI forecasting platform integrating NASA TEMPO, EPA/OpenAQ, and NOAA feeds with 98% prediction accuracy.
- **Top 5 & Rs. 7,000 Prize — Smart Delhi Ideathon 2025:** Civic-tech solution combining backend systems and AI.
- **Winner — Paytm × Nerds Room Hackathon 2025:** Built *CommuneOS*, an AI platform matching members to mentors via resume parsing in under 5s with live personalized dashboards.

Experience

WorkIndulge — Hybrid
IT Development & AI Community Intern — *Sept. 2026 – Present*

- Developed interactive AI-powered learning games and visualization modules using React and FastAPI, enhancing student engagement through adaptive gamification.
- Built full-stack features and high-throughput REST APIs with scalable state management and modular microservices.
- Led applied-AI community initiatives, mentoring students and driving adoption of generative AI workflows.

Cimplifie -Performance Marketing Agency — Remote
Full Stack Development Intern — *Jun. 2026 – Aug. 2026*

- Developed full-stack web applications using React, Spring Boot, and FastAPI, delivering scalable REST APIs, secure JWT auth, and interactive analytics dashboards.
- Engineered automated ingestion pipelines and Selenium worker scripts, cutting manual reporting overhead by **70%**.
- Designed optimized PostgreSQL/MySQL schemas and integrated microservice workflows containerized with Docker.

Projects

Patient Management Service | *Java, Spring Boot, Kafka, gRPC, Docker, PostgreSQL, JUnit* — GitHub

- Engineered 4 independently deployable Spring Boot microservices with zero shared state, utilizing gRPC for low-latency RPC and Apache Kafka for high-throughput event streaming.
- Secured endpoints with Spring Security, JWT, and RBAC; validated core modules using JUnit/Mockito and containerized the system via Docker with Maven builds.

AtmosIQ | *Spring Boot, Spring AI, FastAPI, pgvector, React* — Kaggle | GitHub

- Architected a PM2.5 forecasting and policy intelligence platform pairing a FastAPI ML service (OpenAQ) with a Spring Boot + Spring AI agent grounded in CAQM GRAP policy text; published the **AtmosSim** physics-simulated PM2.5 dataset on Kaggle.
- Built a pgvector RAG pipeline with citation verification, prompt-injection sanitization, and automated safety guardrails.

ResumeRank | *Spring Boot, Next.js, FastAPI, Gemini AI, PostgreSQL, JWT, Docker* — Live | GitHub

- Built an AI ATS resume analyzer that scores resumes against job descriptions and surfaces keyword gaps in seconds.
- Architected a scalable three-tier backend (Next.js, Spring Boot, FastAPI) with JWT authentication and integrated Gemini AI for section-wise rewriting.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript (basic)
Backend & Systems: Spring Boot, Spring AI, Spring Security, Spring Data JPA, Hibernate, FastAPI, Django, REST APIs, Microservices, Apache Kafka, gRPC, WebSockets, Event-Driven Architecture
Databases: PostgreSQL, MySQL, TimescaleDB, Redis, pgvector, Pinecone (Vector DB), Elasticsearch
Frontend: React.js, Next.js, Three.js, WebGL, Zustand, Tailwind CSS, Leaflet
Testing & Build: JUnit, Mockito, Maven, Postman, REST API Testing, Selenium
Developer Tools: Docker, Git, GitHub, CI/CD (GitHub Actions), Vercel
Certifications: Stanford/Coursera — ML Specialization (Andrew Ng) | Telusko/Udemy — Java & Spring Boot Development